Comparison of the Effectiveness Between Conventional and Modified Cinch Suture Techniques in LeFort I Osteotomy

A Systematic Review and Meta-Analysis of Randomized Control Trials

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Background: LeFort I osteotomy changes the morphology of the nose. The cinch suture has been proven to prevent the increase in nasal base and alar width. Different types of cinch sutures have been proposed. However, their effectiveness is unclear.

Aim and Objectives: The aim of this study was to compare the surgical outcomes between conventional and modified cinch techniques through a systematic review and meta-analysis of randomized control trials (RCTs).

Material and Methods: We performed systematic search from Embase, PubMed, and the Cochrane Library according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses statement up to March 2021. The surgical techniques of different cinch sutures were reviewed, and the outcomes of nasal alar width and alar base width were compared between modified and conventional methods.

Results: A total of 4 eligible RCTs were included in this meta-analysis. Pooled data showed no significant difference in alar base width change between modified and conventional methods (mean difference, −0.37; 95% confidence interval, −1.32 to 0.57; $P = 0.44$). Pooled data of 3 studies also showed no significant difference in nasal alar width change (mean difference, −0.11; 95% confidence interval, −1.18 to 0.95; $P = 0.83$).

Conclusion: Based on the current data pooled from the available RCTs, no significant difference was found between the conventional cinch technique and the modified technique.

Key Words: cinch suture, modified cinch, alar base suture, LeFort I osteotomy, orthognathic surgery

(*Ann Plast Surg* 2023;90: S2–S9)

LeFort I osteotomy, the degloving procedure of facial elements from the paranasal periosteum, alters the dimensions of the nasal base, including widening the alar base and nasal alar, nasal tip projection, nasal flaring, and increasing nostril show.[1–5] To mitigate the increase in alar width, the cinch suture has been applied for decades since it was first described by Millard[6] for correcting flat and flaring noses of cleft patients in 1980. The cinch suture technique has been proven to be effective in controlling nasal width and preventing unwanted increases in the alar base caused by the osteotomy.[7,8] In addition, it can restore the normal alar width by avoiding lateral drift of the nasolabial muscle.[9] The effect has been reported to be stable in the medium term from 3 to 12 months postoperatively.[8,10]

However, some surgeons have reported that the conventional cinch technique can lead to problems such as an unequal grasp of soft tissue and unbalanced traction tension on the alae of both sides. Therefore, a modified cinch method was introduced in which a needle is passed though the nasolabial skin fold, engaging the fibroareolar tissue and musculature for a more reliable effect.[11,12] Debate over the surgical outcomes, such as the effectiveness of preventing nasal alar widening between conventional and modified cinch techniques is ongoing. To the best of our knowledge, although some randomized control trials (RCTs) have been conducted,[13–16] no meta-analysis has been performed. Therefore, in this study, we reviewed the literature and performed a meta-analysis of RCTs to compare the surgical outcomes between conventional and modified cinch techniques.

MATERIALS AND METHODS

This systematic review was performed according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) and Assessing the Methodological Quality of Systematic Reviews Guidelines.[17]

Search Strategy and Inclusion Criteria

PubMed, Cochrane Collaboration Central Register of Controlled Clinical Trials, Cochrane Systematic Reviews, and Embase were searched up to March 16, 2021. The following Medical Subject Headings (MeSH) terms were used in combination with Boolean operators (AND, OR, NOT): “Cinch” OR “Cinching suture” OR “Cinch technique” OR “Cinch procedure” OR “Alar Cinch” OR “Alar base Cinch” OR “Alar base suture” OR “cinch suturing” OR “Alar cinching” OR “Alar cinch procedure” AND “Orthognathic surgery” OR “Orthognathic surgical procedures” OR “LeFort I osteotomy” OR “Jaw surgery” OR “maxillofacial surgery” OR “maxillary surgery” OR “mandibular surgery” without limitation of language and year of publication.

Studies where the full text was not available were excluded. Only RCTs that were relevant to conventional and modified cinch suture techniques were retrieved and analyzed. There were no limitations on languages and year of publication. Prospective nonrandomized studies, review articles, retrospective studies, case series, case reports, commentaries, and conference abstracts were excluded. All retrieved studies were required to include the conventional cinch suture technique and the modified technique. The target population comprised patients who received LeFort I osteotomy, either performed as a single procedure or combined with mandible osteotomy in orthognathic surgery.

Three independent authors (J.R.Y., Y.Y.C., and C.H.L.) screened for potentially eligible articles retrieved by the initial search, and

Received October 25, 2022, and accepted for publication, after revision October 25, 2022.

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Conflicts of interest and sources of funding: none declared.

Author Contributions: C.-H.L. participated in the conceptualization. C.-H.L. participated in the methodology. J.-R.Y., Y.-Y.C., C.-H.L. participated in the formal analysis. H.-T.L., C.-H.L. contributed in the resources. J.-R.Y., Y.-Y.C., C.-H.L. participated in the data curation. J.-R.Y., Y.-Y.C. participated in the writing—original draft preparation. J.-R.Y., Y.-Y.C., C.-H.L. participated in the writing—review and editing. H.-T.L., C.-H.L. participated in the visualization. C.-H.L. participated in the supervision. H.-T.L., C.-H.L. participated in the project administration.

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ISSN: 0148-7043/23/9001–00S2

DOI: 10.1097/SAP.0000000000003354

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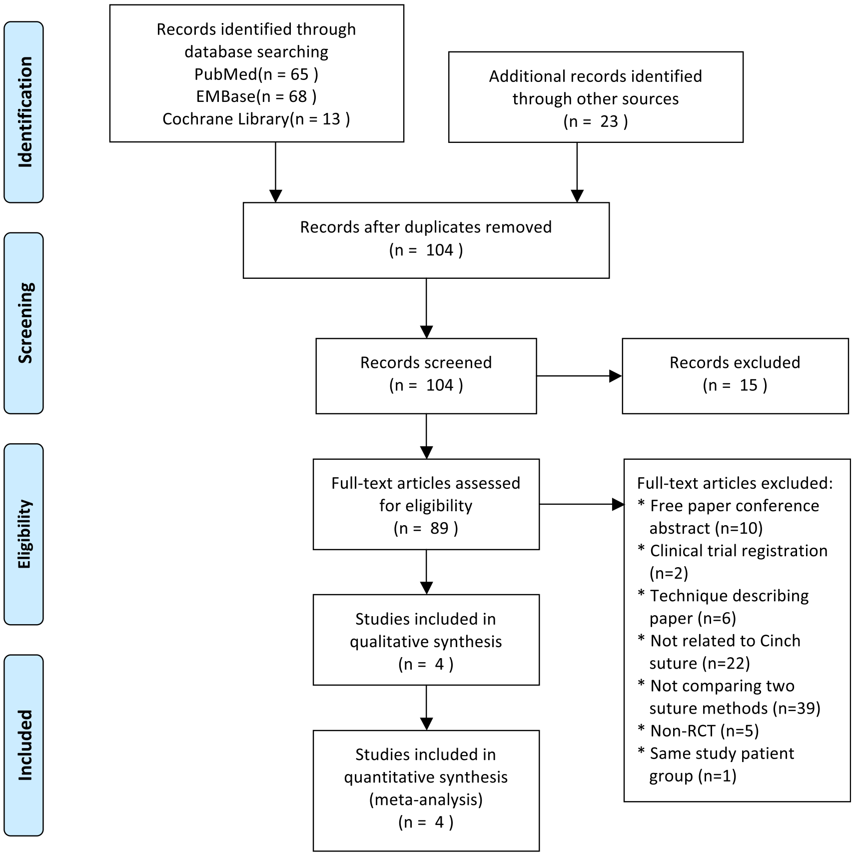

FIGURE 1. PRISMA diagram. The flowchart illustrates the literature search and the process of study selection according to the PRISMA guidelines. full color online

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prospective disagreements were resolved by consensus with all 3 reviewers (J.R.Y., Y.Y.C., and C.H.L.). To prevent missing any relevant studies, the references for the previously included studies and published systematic reviews were also manually searched. If there were duplicate publications or patient groups, only the most recent study with more complete patient data was included.

Data Extraction and Quality Assessment

After reviewing the full texts of the studies identified in the search, 3 independent authors (J.R.Y., Y.Y.C., and C.H.L.) extracted data and crosschecked the eligible studies. Inter-reviewer discrepancies in the selection of articles and the extraction of data were resolved by discussion and the consensus of all 3 reviewers. The extracted data included author name, year of publication, study design, patients' demographics (age, sex), mean follow-up time, direction of maxillary movement, different cinching surgical procedures, and long-term outcome parameters (nasal alar width, alar base width). The quality of the included studies was assessed using version 2 of the Cochrane risk-of-bias tool (RoB 2.0).[18]

Statistical Analysis

Outcome measurements (mean and standard deviation values) were extracted for continuous variables in the conventional and modified cinch techniques and were defined as differences in the widths of the nasal alar and alar base preoperatively and postoperatively. If necessary and available, variables that were needed for meta-analysis were retrieved and calculated based on the data extracted from the eligible articles. RevMan software 5.3 (The Cochrane Collaboration) was used to perform statistical analysis, and P values <0.05 were regarded as being statistically significant. The I^2 test was used to determine between-trial heterogeneity. Random effects models were used to calculate pooled estimates of the mean differences to take potential interstudy heterogeneity into account.

RESULTS

Literature Search and Selection

A total of 146 articles were retrieved in the initial search (65 from PubMed, 68 from Embase, and 13 from the Cochrane Library), and 23 additional articles were identified through other resources. After removal of duplicates, 104 articles were screened. Fifteen articles were excluded because of irrelevant titles and contents, and the remaining 89 articles underwent full-text review to assess their eligibility. Among them, 10 articles were free article abstracts of conferences, 2 were clinical trial registrations without attributable results, 6 described techniques, 22 were not related to the cinch technique, 39 did not compare different cinch suture methods, 5 were not RCTs, and 1[19] shared the same study patient group with one of the eligible articles.[15] Finally, a total of 4 studies[13–16] were reviewed as eligible articles for inclusion in the meta-analysis. Figure 1 shows the PRISMA diagram of the selection process. The outcomes and characteristics of the eligible articles are shown (Table 1).

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TABLE 1. Characteristics of the Included Studies

Study ID	Included Patients (No. Patients)	Mean Age (Range), Year	Overall Male/Female Ratio	Modified Alar Cinch Suture (No. Patients)	Classic Alar Cinch Suture (No. Patients)	Reported Outcome	Postoperative (mo)	Inclusion Criteria	Exclusion Criteria	Upper Jaw Movement
Rauso 2010	40	Conventional: 27 (19–34) Modified: 25 (18–31)	17:23	Modified alar cinch suture (20)	Classic alar cinch suture (20)	L-M: Distances (mm) between the left alar base and the columella M-R: Distances (mm) between the Columella and the right alar base	6	(1) Skeletal class III facial deformity (2). All the patients had bimaxillary operations, with or without genioplasty	Cleft lip, previous nasal operation, and previous or simultaneous additional midfacial operations	Maxillary advancement or maxillary impaction or maxillary clockwise rotation or maxillary counterclockwise rotation
Ritto 2011	38*	All: 25.9 (17–50)	17:21	Extra oral alar base cinch suture (18†)	Classic intraoral suture (15)	Alar width Alar base width Maxillary movements	>3	(1) LeFort I osteotomy in 1 segment; (2) maxillary advancement of ≥3 mm, or maxillary superior repositioning of ≥3 mm using the central incisor as the reference	Clefts, growth, history of facial trauma with fractures of facial bones, history of surgical procedures in the nasal region, facial asymmetries, postoperative complications	Maxillary advancement of ≥3 mm, or maxillary superior repositioning of ≥3 mm using the central incisor as the reference

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Chen 2015	48	Conventional: 23.78 (18–34) Modified: 24.13 (19–57)	15:33	Modified alar base cinch suture (24)	Classic alar base cinch suture (24)	Inter-canthal distance Nasal height Nasal length Nasal tip protrusion Nasal width Alar base width Right nostril show vertical dimension Left nostril show vertical dimension Columellar length NLA Cutaneous height of upper lip Overall upper lip height Vermilion height of upper lip Lower prolabial width Upper lip protrusion	6	LeFort I osteotomy	Cleft of the lip or palate, dentofacial trauma, or previous nasal septum or nasal tip operations	—
Wang 2015	79‡	All: 23.2 ± 3.4	22:57	Extraoral alar base cinch suture (17)	Modified intraoral alar base cinch suture (31)	Sbal-Sbal Al-Al G. lat-G. lat	3 and 6	LeFort I osteotomy	Growth, age >35, previous nasal operations, nasal deformities, facial asymmetries	Maxillary setback, unchanged, and forward

*38 patients fitted the inclusion criteria initially. One patient (17) was excluded due to a postoperative infection, and another 2 patients (22 and 29) did not return for final photographs. Therefore, 35 patients were allocated to group 1 (classic intraoral suture) and group 2 (extra oral alar base cinch suture) eventually.

†20 patients were allocated to group 2 (extra oral alar base cinch suture) initially. Only 18 patients were included eventually because 2 patients from group 2 (30 and 31) presented a large disparity of results, probably owing to errors during photo acquisition, editing, and measuring. Aiming to avoid miscalculations, these patients were not considered in the final calculations.

‡79 Patients were randomized into 3 groups in this study, including group 1 (31) with traditional intraoral alar base cinch suture, group 2 (17) with extraoral alar base cinch suture and group 3 (31) with modified intraoral alar base cinch suture. We chose group 2 and group 3 as our target groups.

NLA, nasolabial angle; Sbal, subalare; Al, alare (the most lateral point on the nasal ala); G. lat, the lateral point of the nasal alar groove.

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Study Characteristics

The 4 eligible studies were published from 2010 through 2015, which included and analyzed a total of 205 patients (Table 1) who received LeFort I osteotomy with the cinch suture from January 2006 through March 2014. Two different cinch techniques were described and compared: the conventional method (90 patients) and the modified method (79 patients).

Surgical Procedures

Conventional Technique

We defined the conventional cinch suture as the application of extraoral pressure on the alar base region and grasping the nasalis muscle transorally without dermis involvement[15] (Fig. 2A). Although not using the same terminology or surgical procedures and materials, these 4 eligible RCTs shared the same concept of this conventional technique. Rauso et al[13] described it as a "classic method" using a 3/0 nonabsorbable suture to anchor the fibroareolar tissues under both alae. With a finger pressing on the ala extraorally, the fibroareolar tissue was identified. Tissue movement was observed intraorally through the vestibular incision. The fibroareolar tissue was anchored appropriately with confirmation by pulling the cinch sutures and making sure that bilateral alae moved similarly to the medial site. A hole was made in the anterior nasal spine (ANS) with 2 free ends of the sutures passing through and making a knot.

Ritto et al.[14] also described this conventional technique as a "classic method" that is retracting the upper lip with the thumb and used the index finger to create extraoral pressure on the alar base. Dentate forceps were then used to grasp the tissue through the intraoral incision, and a nonresorbable monofilament suture was passed through the tissue that was held by the forceps. The same procedure was done to the contralateral side. The suture was tightened with adequate tension on the alar base, and the intraoral incision was then closed.

In Chen et al's study,[15] the conventional technique of cinch began from the bilateral nasalis muscle alar part and went through a drilled hole in the ANS with 3-0 nylon suture, whereas the technique that Wang et al.[16] termed as "traditional" method started from the upper buccal gingival incision. Dentate forceps were used to grasp the nasalis muscle at the most lateral part (Al) with an adequate nasalis response and passed through the silk suture. The 2 free ends of the sutures were then tied at the midline of the nose (Table 2).

Modified Technique

The modified cinch technique was defined as the conventional cinch suture with additional dermis involvement (Fig. 2B). Rauso et al's[13] modified technique involved the insertion of an 18-gauge needle through the skin point of the right alar base, then passing it through the fibroareolar tissue. A 3/0 nonabsorbable suture was then inserted through the needle intraorally to extraorally. The needle was withdrawn through the right alar base without leaving the skin point, and then went back to the oral cavity in a medial position. Ultimately, the needle was withdrawn from the right alar base with the suture placed through the soft tissues. The same procedure was done on the contralateral side. Finally, the 2 free ends of the sutures were tied after passing through a hole made in the ANS.

Ritto et al[14] used a thick needle with a nonresorbable monofilament suture to catch the alar base tissue from the intraoral incision to the skin alar-facial junction extraorally. The needle was then reintroduced into the intraoral incision via the same puncture site. The direction of the needle deviated after insertion into the skin and before appearing in the intraoral incision. The alar base tissue was grasped by the cinch suture and remained beneath the skin. The suture was tightened with adequate tension and alar base response, and the upper buccal gingival incision was closed.

Chen et al[15] started the suture with 3-0 nylon from the bilateral alar part of the nasalis muscle, and the dermis tissue over the alar base was also included. The suture went through a drilled hole in the ANS and made a knot. In Wang et al's method,[16] an 18-gauge needle with double strand steel wire was inserted through the skin point located in the lower 1/3 from the most lateral part of the nasalis muscle to the alar base into the oral cavity. The double strand steel wire was withdrawn with silk to subcutaneous tissue and reinserted into the oral cavity in a medial position, meeting the previous needle insertion at a 30- to 45-degree angle. The silk from the steel wire was retracted and the needle was withdrawn. The nasalis muscle was then tightened with attention to alar base response. The same procedure was done at the contralateral side. The 2 free ends of the sutures were then tied after passing through a hole made in the ANS (Table 2).

Outcomes

A variety of outcome parameters were described in the eligible studies, including alar base width, nasal alar width, facial-alar junction width, maxillary movement, nasal height, length, tip projection, and changes in lip parameters.

Pooled Nasal Alar Width Differences

Three eligible studies reported nasal alar changes preoperatively and postoperatively. There were no significant differences between the conventional group and modified group at 3 months of follow-up (mean difference, −0.09; 95% confidence interval [95% CI], −1.71 to 1.52;

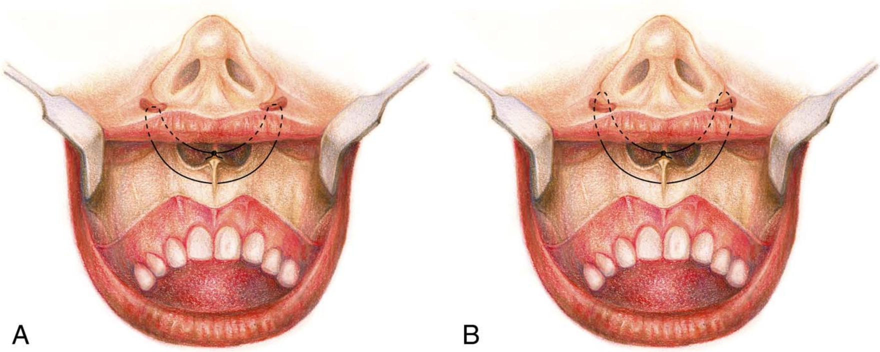

FIGURE 2. Comparison between conventional cinch and modified cinch suture methods: The schematic diagram demonstrates the concept of cinch technique with or without the dermis grasping. A, The conventional cinch suture was achieved by applying extraoral pressure on the alar base region and grasping the alar part of nasalis muscle transorally or extraorally. B, The modified cinch suture with additional dermis involvement. full color online

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TABLE 2. Suturing Methods Used in the Included Studies

Classic alar cinch suture	Rauso 2010	A 3/0 nonabsorbable suture was used to anchor the fibroareolar tissues under both alae. The fibroareolar tissue was identified by applying pressure on the ala of the nose extraorally and observing tissue movement intraorally through the vestibular incision. The correct anchoring of the fibroareolar tissue was confirmed by applying tension on the cinch sutures and making sure that medial movements of both alae were similar. The 2 free ends of the sutures were then passed through a hole made in the nasal spine, making a knot.
	Ritto 2011	Although retracting the superior lip with the thumb, the index finger was used to apply extraoral pressure on the alar base region and dentate forceps grasped this tissue through the intraoral incision. The lip was then released and a nonresorbable monofilament suture was passed through the tissue previously held by the forceps. The same procedure was applied on the opposite side. After passing the suture on both sides, it was tightened with attention to the alar base response. If this response was not judged to be adequate, the suture was released and the procedure restarted from the beginning. If the alar base suture was judged to be adequate, the vestibular incision was then closed in a routine fashion, without performing a V-Y lip closure.
	Chen 2015	The suture began from the bilateral alar part of the nasalis muscle and passed through a hole drilled in the ANS.
	Wang 2015	From the vestibular incision, the author applied dentate forceps to grasp the nasalis muscle corresponding to Al with attention to the nasalis response and pass through the suture with silk. The same procedure was done through opposite side. The 2 free ends of the sutures were made a knot in the midline of nose.
Modified alar cinch suture	Rauso 2010	An 18-gauge needle was inserted through the skin-point R and exited through the fibroareolar tissue. A 3/0 nonabsorbable suture without a needle was inserted through the needle from the oral cavity to the outside. The needle was retracted through point R without leaving the skin point, then 4to the oral cavity in a medial position. Finally, the needle was retracted from point R leaving the suture through the soft tissues. The same procedure was done through point L. The 2 free ends of the sutures were then passed through a hole made in the nasal spine making a knot.
	Ritto 2011	A nonresorbable monofilament suture with a thick needle was used to transect the alar base tissue from the intraoral incision to the skin extraorally, exiting at the alar-facial junction. The needle was then reinserted into the intraoral incision through the same puncture site (this is why this technique is better performed with a thick needle, so the needle is easily inserted back in the same hole of previous exit). After passing the skin and before exiting in the intraoral incision, the direction of the needle was changed. In this way, the suture was able to grasp the alar base tissues and remain beneath the skin. After passing the suture on both sides, the lip was released and the suture tightened with attention to the alar base response. If this response was not judged to be adequate, the suture was released and the procedure restarted from the beginning. If the alar base suture was judged to be adequate, the vestibular incision was then closed in a routine fashion, without performing a V-Y lip closure.
	Chen 2015	The suture began from the bilateral alar part of the nasalis muscle and dermis tissue over the alar base, and then passed through a hole drilled in the ANS.
	Wang 2015	An 18-gauge needle with double strand steel wire was inserted through the skin point located at the lower 1/3 from Al to Sbal into the oral cavity. Double strand steel wire was withdrawn with silk to subcutaneous tissue and reinserted into the oral cavity in a medial position, meeting the previous needle insertion at a 30–45° angle. The silk was retracted from the steel wire and the needle was withdrawn. The muscle was tightened paying attention to the alar base response. The same procedure was done through the opposite side. The 2 free ends of the sutures were then passed through a hole made in the nasal spine making a knot.

ANS, anterior nasal spine; Al, alare (the most lateral point on the nasal ala); Point R, nasofacial skin fold at the right alar base; Point L, nasofacial skin fold at the left alar base; Sbal, subalare.

$P = 0.91$; heterogeneity, $I^2 = 90\%$) (Fig. 3A) or at 6 months of follow-up (mean difference, 0.36; 95% CI, −0.54 to 1.27; $P = 0.43$; heterogeneity, $I^2 = 52\%$) (Fig. 3B). Pooled data of all studies also showed no significant difference after more than 3 months of follow-up (mean difference, −0.11; 95% CI, −1.18 to 0.95; $P = 0.83$; heterogeneity, $I^2 = 81\%$) (Fig. 3C).

Pooled Alar Base Width Differences

Alar base width differences were reported in 4 studies. There were no significant differences between the conventional group and modified group at 3 months of follow-up (mean difference: −0.38; [95% CI, −1.82-1.07]; $P = 0.61$; heterogeneity, $I^2 = 87\%$) (Fig. 3D) or at 6 months of follow-up (mean difference: −0.10; [95% CI, −1.25-1.05]; $P = 0.87$; heterogeneity, $I^2 = 78\%$) (Fig. 3E). Pooled data of all studies also showed no significant difference after more than 3 months of follow-up (mean difference: −0.37; [95% CI, −1.32-0.57]; $P = 0.44$; heterogeneity, $I^2 = 78\%$) (Fig. 3F).

Compliance and Safety

No reported complications were observed in the included studies.

Study Quality Assessment

The quality of the RCTs that we included was assessed using version 2 of the Cochrane risk-of-bias tool (RoB 2.0).[18] Overall, the 4 RCTs were classified overall risk as high or some concerns. The main causes of risks were caused by confounding factors in the randomization process in Chen et al's,[15] Rauso et al's[13] and Ritto et al's[14] studies, which were ranked as "some concerns." Rauso et al's[13] and Wang et al's[16] studies showed high risk of bias in measurement of outcome, whereas Ritto et al's[14] study showed some concerns. All the articles presented low risk of bias in the items of deviations from intended interventions, missing outcome data, and selection of the reported result. The risk of bias in these studies is summarized in Figure 4.

DISCUSSION

Changes in nasal morphology including alar base widening, nasal tip upturning, increasing nasolabial angle and nostril show[20] usually occur after LeFort I osteotomy as the maxilla moves upward and forward.[21] To overcome these issues, the alar base cinch technique was proposed.[1,6] This study is the first meta-analysis of RCTs to provide the latest evidence that the modified cinch suture technique is not significantly better than the conventional technique.

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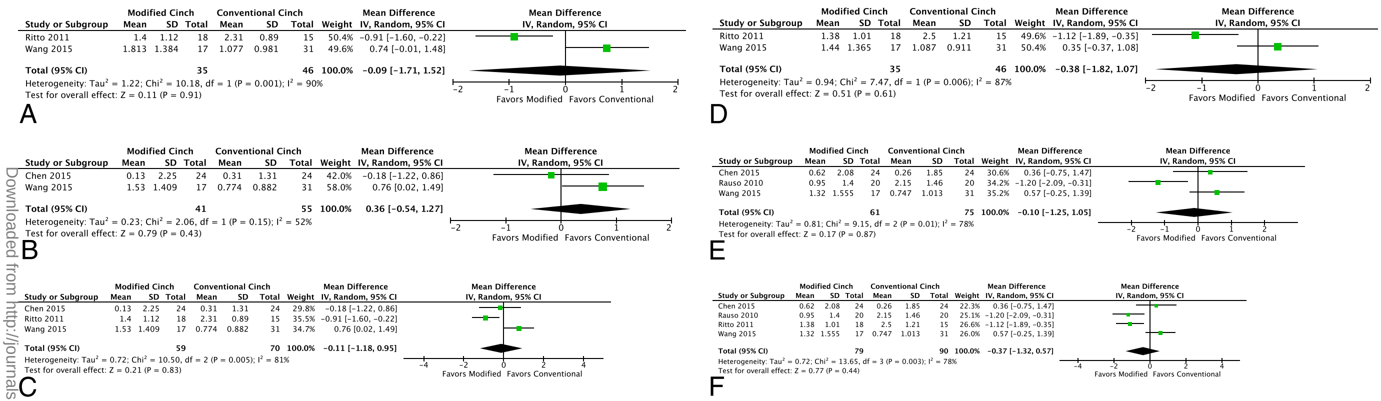

FIGURE 3. Forest plots of the primary outcomes. A, Forest plot of included studies evaluating the effects on nasal width after LeFort I osteotomy at 3 months follow-up after surgery using a random effect model. B, Forest plot of included studies evaluating the effects on nasal width after LeFort I osteotomy at 6 months follow-up after surgery using a random effect model. C, Forest plot of included studies evaluating the effects on nasal width after LeFort I osteotomy after more than 3 months follow-up after surgery using a random effects model. D, Forest plot of included studies evaluating the effects on alar base after LeFort I osteotomy at 3 months follow-up after surgery using a random effects model. E, Forest plot of included studies evaluating the effects on alar base after LeFort I osteotomy at 6 months follow-up after surgery using a random effects model. F, Forest plot of included studies evaluating the effects on alar base after LeFort I osteotomy after more than 3 months follow-up after surgery using a random-effects model. full color online

Traditionally, the cinch suture is achieved by applying extraoral pressure on the alar base region and grasping the nasalis muscle transorally. The alar base tissue can be moved medially 4 to 6 mm without undue tension. A 2-O or 3-O nonabsorbable suture is then passed from one "alar base" to the other, and they are pulled together. Adequate tension and symmetrical movement of both sides should be confirmed to prevent asymmetry.[1] Howley et al[22] reported that the conventional cinch suture could slightly decrease alar base widening, but that the difference between the control and cinch groups was not statistically significant. Modified cinch suture techniques were then proposed in several studies, either with dermis involvement[12,14,16,19] or a transeptal approach.[23] In 2010, Rauso et al[13] reported that a modified technique involving the dermis and nasalis muscle was more stable than the conventional technique, and in 2011, Ritto et al[14] reported similar results.

Liu et al[24] reported a systematic review that compared modified and conventional alar base cinch suture techniques in 2014 and concluded that the modified cinch technique was more effective than the conventional technique in maintaining nasal alar and alar base width. However, they only included 3 eligible articles (Ritto et al,[14] Rauso et al,[13] and Nirvikalpa et al[23]), and there was no pooled data for meta-analysis. Moreover, the modified technique described by Ritto et al[14] and Rauso et al[13] involved simultaneously grasping the dermis and nasalis muscle, whereas the modified technique described by Nirvikalpa et al[23] involved transseptal sutures. The cinch methods that have been compared in Nirvikalpa et al's[23] study focused on an anchoring bite taken through the nasal septum behind its anterior edge or not. This concept of modification was different from the other RCT studies; therefore, we did not recruit it in for meta-analysis. The results from the risk of bias assessment also illustrated differences in judgment. Liu et al[24] used RoB 1.0 to rank the eligible studies. However, we used RoB 2.0 to perform the evaluation progressively. They ranked the random sequence generation in Ritto et al[14] article as high risk, whereas we ranked it as some concerns because of randomization process. The measurement of outcome was found lacking of relative information, making the blinding of outcome assessment high risk in Rauso et al[13] and Wang et al[16] studies. Since then, the overall risk of bias was high in Rauso et al[13] and Wang et al[16] study according to RoB 2.0 tool assessment.

More recently, in 2015, Wang et al[16] and Chen et al's[15] studies reported no significant differences between modified (including the dermis and nasalis muscle) and conventional (including the nasalis muscle only) methods. These 2 studies recruited more cases and compared more parameters in detail. Therefore, we enrolled these 2 articles and unified the definition of the modified technique as involving the dermis, and extracted data of preoperative and postoperative nasal ala

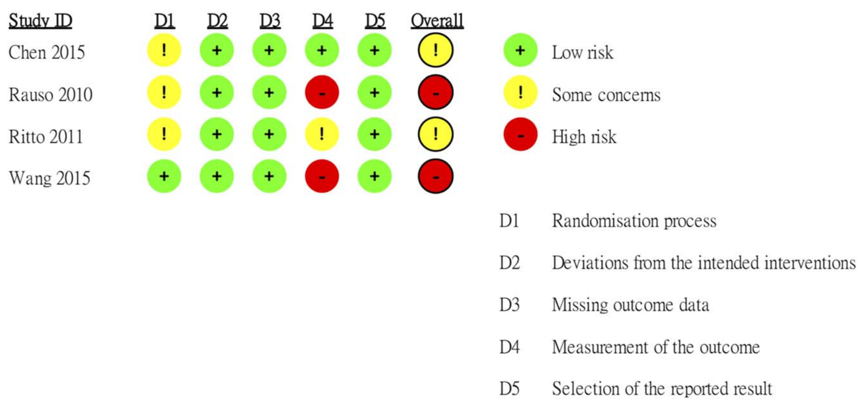

FIGURE 4. Risk-of-bias analysis of the included studies. Risk-of-bias summary: The judgment of the authors regarding each risk-of-bias item for the included studies. full color online

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and alar base widths for meta-analysis. The results showed no significant differences between the 2 methods in either nasal ala or alar base width changes, which is different from previous systematic reviews.

The timing of evaluation is also a controversial issue. Edler et al[25] reported no difference in alar width between 3 to 6 months and 9 to 12 months after surgery. However, Ritto et al[14] suggested that the timing should depend on the region to be evaluated and that the alar base region should be evaluated 4 months postoperatively. Osher et al[7] reported that reductions in alar width became stable at 1 year of follow-up. In our study, the 4 eligible articles assessed outcomes at 3 to 6 months postoperatively. Nonetheless, subgroup analysis of either 3 or 6 months revealed no significant difference between the modified and conventional groups. Our study results may indicate that the intrinsic part of the cinch technique is grasping the alar nasalis muscle. To avoid alar widening after LeFort I osteotomy, the crucial factors are that the muscle structure of bilateral alae should be gathered up and the degloved soft tissue must be reattached to the bone. If the tension of muscle grasping and tightening is appropriate, the cinch effect will be optimal after the degloved musculoperiosteal flap has been repositioned and becomes stable after several months, regardless of the dermis involvement.

There are several limitations to this study. First, the detailed procedures of the cinch technique varied from institution to institution. For example, Chen et al's modified cinch method involved grasping the nasalis and dermis via an intraoral incision, whereas the other authors introduced the needle extraorally. Second, although all the suture stitches mentioned in the articles were nonabsorbable, the tensile strength of the suture materials still differed. Chen et al used the single-stranded 3-0 nylon suture, whereas Wang et al used multistrand silk. The other studies did not specify the suture materials or caliber. This may have affected the postoperative duration of nasal width control. Third, not every eligible study controlled tightness when tying the suture knot. This may have affected the tension of each cinch suture and how long it lasted. Fourth, care should be taken as nasal width widening may be more severe in patients with maxillary advancement or clockwise rotation sagittally. None of the 4 eligible studies controlled the amount of maxillary movement as a parameter. Fifth, care should be taken that the ethnic variations may cause higher heterogeneity because the midface structure and skin elasticity are different from Asian and *Caucasian* races. The 4 RCTs that we have recruited for meta-analysis included studies from Taiwan, China, Italy, and Brazil. Given that there are ethnic variations in nasal morphology, the findings of this study should be applied cautiously. Furthermore, subgroup analysis is required and expected if more RCTs are conducted and published in the future. All of these limitations may have caused higher heterogeneity in the meta-analysis. Ultimately, all eligible studies enrolled limited case numbers and had relatively short follow-up periods of no more than 6 months. Further RCTs with more patients and longer follow-up are needed with a lower risk of bias to confirm our findings.

CONCLUSION

Based on the current data pooled from the available RCTs, no significant difference was found between the conventional cinch technique and the modified technique. Long-term results and more RCTs are needed to verify changes in nasal width between different techniques after LeFort I osteotomy.

REFERENCES

1. Collins PC, Epker BN. The alar base cinch: a technique for prevention of alar base flaring secondary to maxillary surgery. *Oral Surg Oral Med Oral Pathol*. 1982;53: 549–553.
2. Sasaki M, Izumi K, Ikezaki S, et al. Nasal morphologic changes after Le Fort I osteotomy. *Int J Oral Maxillofac Surg*. 2015;44:e135.
3. Westermark AH, Bystedt H, von Konow L, et al. Nasolabial morphology after Le Fort I osteotomies effect of alar base suture. *Int J Oral Maxillofac Surg*. 1991;20: 25–30.
4. Jensen A. Soft tissue changes associated with double jaw surgery. *Am J Orthod Denfac Orthop*. 1992;101:265–275.
5. Betts NJ, Vig K, Vig P, et al. Changes in the nasal and labial soft tissues after surgical repositioning of the maxilla. *Int J Adult Orthodon Orthognath Surg*. 1993;8:7–23.
6. Millard DR Jr. The alar cinch in the flat, flaring nose. *Plast Reconstr Surg*. 1980; 65:669–672.
7. Osher J, Witherow H. Long term stability of the cinch suture. *Br J Oral Maxillofac Surg*. 2011;49:S88.
8. Moroi A, Ishihara Y, Sotobori M, et al. Evaluation of soft tissue morphologic changes after using the alar base cinch suture in Le Fort I osteotomy in mandibular prognathism with and without asymmetry. *J Craniomaxillofac Surg*. 2014;42: 718–724.
9. Mustafa K, Shehzana F, Bhat HH. Assessment of alar flare and efficacy of alar cinch suture in the management of alar flare following Le Fort 1 superior repositioning: a comparative study. *J Maxillofac Oral Surg*. 2016;15:528–534.
10. Stewart A, Edler RJ. Efficacy and stability of the alar base cinch suture. *Br J Oral Maxillofac Surg*. 2011;49:623–626.
11. Shams MG, Motamedi MHK. A more effective alar cinch technique. *J Oral Maxillofac Surg*. 2002;60:712–715.
12. Rauso R, Gherardini G, Tartaro G, et al. A modified alar cinch suture technique. *Eur J Plast Surg*. 2009;32:341–344.
13. Rauso R, Gherardini G, Santillo V, et al. Comparison of two techniques of cinch suturing to avoid widening of the base of the nose after Le Fort I osteotomy. *Br J Oral Maxillofac Surg*. 2010;48:356–359.
14. Ritto FG, Medeiros PJ, de Moraes M, et al. Comparative analysis of two different alar base sutures after Le Fort I osteotomy: randomized double-blind controlled trial. *Oral Surg Oral Med Oral Pathol Oral Radiol Endod*. 2011;111:181–189.
15. Chen CYH, Lin CCH, Ko EWC. Effects of two alar base suture techniques suture techniques on nasolabial changes after bimaxillary orthognathic surgery in Taiwanese patients with class III malocclusions. *Int J Oral Maxillofac Surg*. 2015; 44:816–822.
16. Wang ZQ, Wang XX, Li ZL, et al. Comparison of three surgical techniques for controlling nasal width after Le Fort I osteotomy. *Beijing Da Xue Xue Bao Yi Xue Ban*. 2015;47:104–108.
17. Moher D, Liberati A, Tetzlaff J, et al, PRISMA Group. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *Ann Intern Med*. 2009;151:264–269.
18. Higgins JP, Altman DG, Gøtzsche PC, et al. The Cochrane Collaboration's tool for assessing risk of bias in randomised trials. *BMJ*. 2011;343:d5928.
19. Chen Y, Lin CH, Ko EC. Nasolabial changes affected by 2 different alar base cinch suture techniques after maxillary LeFort I osteotomy in class III malocclusions: randomized controlled trial. *Cleft Palate-Craniofacial Journal*. 2014; 51:e19.
20. Worasakwutiphong S, Chuang Y-F, Chang H-W, et al. Nasal changes after orthognathic surgery for patients with prognathism and class III malocclusion: analysis using three-dimensional photogrammetry. *J Formos Med Assoc*. 2015; 114:112–123.
21. Park S-B, Yoon J-K, Kim Y-I, et al. The evaluation of the nasal morphologic changes after bimaxillary surgery in skeletal class III malocclusion by using the superimposition of cone-beam computed tomography (CBCT) volumes. *J Craniomaxillofac Surg*. 2012;40:e87–e92.
22. Howley C, Ali N, Lee R, et al. Use of the alar base cinch suture in Le Fort I osteotomy: is it effective? *Br J Oral Maxillofac Surg*. 2011;49:127–130.
23. Nirvikalpa N, Narayanan V, Wahab A, et al. Comparison between the classical and a modified trans-septal technique of alar cinching for Le Fort I osteotomies: a prospective randomized controlled trial. *Int J Oral Maxillofac Surg*. 2013;42: 49–54.
24. Liu X, Zhu S, Hu J. Modified versus classic alar base sutures after LeFort i osteotomy: a systematic review. *Oral Surg Oral Med Oral Pathol Oral Radiol*. 2014;117:37–44.
25. Edler RJ, Wertheim D, Greenhill D, et al. Quantitative use of photography in orthognathic outcome assessment. *Br J Oral Maxillofac Surg*. 2011;49:121–126.

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